SKYFORCE Interceptor: Final Resolution of Remaining Simulation Issues

Team 08, Anti-Drone Interceptor, Senior Track. Prepared 10 October 2026. Answers the four items left open in the guide-review resolution: forward-flight propulsion margin, complete-mission energy, physical capture/recovery, and the illustrative safe-area boundaries.

This document pushes each item as far as calculation honestly allows. Two of the four (propulsion margin, mission energy) now have a defensible quantitative answer. The other two (capture/recovery performance, and a real site boundary) cannot be produced by any further calculation — they need either a dedicated deployment-dynamics model, physical testing, or an actual site, none of which exist yet. Those two are addressed by stating plainly what is and is not established, and what would be needed to close them. Presenting invented numbers for either would be worse than leaving them open.

1. Propulsion performance in forward flight

The static thrust margin already established (2.27x at 18 kg, 1.79x at 23.5 kg, both at 25 m/s) uses the motor's published static bench data, which has no forward airspeed component. A real propeller loses some thrust capability as forward speed rises (the advance-ratio effect), so the open question was whether that loss could erase the margin.

No measured thrust-versus-airspeed curve exists for the G28x9.2 CF propeller, so an exact number cannot be produced without a wind-tunnel or flight test. What can be done honestly is a sensitivity check: apply a typical forward-flight thrust de-rating range reported for multirotor propellers operating well below their pitch-speed limit (15 to 30 percent is a commonly cited range for this class of propeller at these speeds, and is treated here as an assumed bound, not a measured one), and check whether the margin survives it.

Assumed de-rating	Available thrust	Margin at 18.0 kg, 25 m/s	Margin at 23.5 kg, 25 m/s
15% (optimistic)	365.2 N	1.93x	1.52x
20% (mid)	343.7 N	1.81x	1.43x
30% (conservative)	300.8 N	1.59x	1.25x

Even under the conservative 30% assumption, margin stays above 1.0 at both masses, meaning 25 m/s remains achievable in principle even allowing for a forward-flight thrust loss well outside the optimistic case. The 23.5 kg (post-capture) margin is tighter (1.25x in the worst case), but this does not threaten the current design, because the tow phase was already planned at near-hover speed rather than a 25 m/s dash (the tow-phase energy estimate in the sizing document scales hover power by mass, not sprint power) — so the aircraft is never actually asked to sprint at the heavier post-capture mass in the current mission plan.

Resolution: the propulsion margin is quantitatively defensible under a reasonable range of forward-flight de-rating assumptions, and the mission plan already avoids the one case (high-speed flight while loaded) where the margin would be tightest. The de-rating percentages themselves remain an assumed range, not measured data, and should be labeled as such if quoted in the report.

2. Complete-mission energy consumption

The earlier sizing document gives a budget ceiling (207 Wh no-tow, 253 Wh with tow) built from a generic 120 s sprint plus 120 s hover-equivalent assumption. The MATLAB simulation separately reports actual energy for specific geometric scenarios (33.30 Wh attack, 36.84 Wh defense, 95.68 Wh tow), which are shorter than the generic budget profile and therefore smaller. Neither of these is a complete-mission total: the budget is a sizing ceiling, and the simulated figures cover only the modeled flight phase, not pre-launch, landing, or avionics draw.

A complete-mission estimate can be built by taking the simulation's actual (worst-case) numbers and adding a stated buffer for the phases the trajectory model does not track:

Added phase	Assumption	No-tow case	Tow case
Launch spin-up	5 s at full sprint power (3.61 kW)	5.0 Wh	5.0 Wh
Post-release / landing descent	10 s at nominal hover power (no-tow); 20 s at loaded hover power (tow, settling before release)	4.3 Wh	12.9 Wh

Added phase	Assumption	No-tow case	Tow case
Avionics and radio (continuous)	~30 W over the flight duration	0.3 Wh	0.8 Wh
Buffer subtotal		9.7 Wh	18.8 Wh

```
No-tow complete-mission estimate (worst case = defense, 36.84 Wh)
   36.84 + 9.7  = 46.5 Wh   against a 207 Wh ceiling  (about 22% of budget used)

Tow complete-mission estimate (95.68 Wh)
   95.68 + 18.8 = 114.4 Wh  against a 253 Wh ceiling  (about 45% of budget used)
```

Resolution: using the simulation's own worst-case phase numbers plus a stated, itemised buffer for the untracked phases, the complete mission is estimated at about 47 Wh (no-tow) and 114 Wh (tow) — both comfortably inside the sizing ceilings, with roughly 78% and 55% headroom respectively. This is a bound built from real simulated numbers plus named assumptions, not a new simulation run; it shows the budget has large margin rather than proving the exact draw of a specific flight.

3. Physical capture and recovery

This item cannot be resolved by further calculation, and no attempt is made here to invent a capture-probability number or a net-deployment dynamics result. A point-mass trajectory simulation has no way to represent the net opening, the mesh fouling the target's rotors, or the parachute's opening shock and canopy behaviour — these are different physics entirely (structural, fluid-dynamic and mechanism-timing problems), not an extension of the guidance and energy model already built.

What is already established, and what is not:

- Established (sizing document): net opening diameter (4 m, from a geometric coverage rule of three times an assumed 1.2 m target span), parachute canopy diameter (2.5 m) and the resulting descent rate (4.74 m/s) for a 5.5 kg load, from a straightforward drag-equation sizing calculation.
- Not established, by any document in this project: whether the net reliably fouls the target's propellers once fired, what fraction of attempts would miss or fail to disable the target, the parachute's opening shock loads, or how the captured pair behaves if it swings or tumbles before the canopy fully opens.

Resolution: present the net and parachute sizing as design-target geometry (already done), and state the capture and deployment performance explicitly as unvalidated — outside the scope of this trajectory model. The two realistic paths to closing this, neither available in the remaining hackathon time, are a dedicated deployment-dynamics simulation, or physical bench and drop testing of the net and parachute. Recommend the report say this plainly rather than imply a number that does not exist.

4. Illustrative keep-out and safe-area boundaries

No real site has been specified at any point in this project, so the 300 m keep-out radius and 120 m safe-area radius remain scenario placeholders, not approved limits. What can be added is a grounding reference, to show the placeholders are at least plausible rather than arbitrary.

India's drone regulator (DGCA), under Civil Aviation Requirements Section 3, Series X, Part I, sets a much larger outer restriction: no drone flight is permitted within 5 km of the perimeter of the six busiest airports (Delhi, Mumbai, Chennai, Kolkata, Bengaluru, Hyderabad) and within 3 km of any other civil, private or defence airport, without clearance.

This is a different concept from the simulation's keep-out boundary, and the two should not be conflated: the DGCA distance is the outer no-fly zone within which a rogue drone is already flying illegally (which is exactly why interception is being proposed in the first place). The simulation's 300 m "keep-out" boundary is a tighter, tactical limit inside that outer zone — the point past which the engagement would be considered too close to a sensitive area (runway, terminal, crowd) to continue safely, not the legal no-fly boundary itself. The engagement radius already assumed (1.5 to 2 km) sits comfortably inside the DGCA's 3 to 5 km outer zone, which is consistent: a real intercept would be completed with room to spare before reaching the regulator's own boundary.

Resolution: keep 300 m and 120 m explicitly labeled as illustrative tactical sub-boundaries for the demonstration, now grounded by the DGCA reference above rather than picked arbitrarily. They remain scenario assumptions, not approved operational limits, and should be presented as such until an actual site is given.

Source: DGCA Civil Aviation Requirements, Section 3, Series X, Part I, Para 13.1 (effective 1 December 2018), as reproduced in parliamentary answers and public summaries of the Requirements.

5. Summary for the next simulation pass

Item	Status after this document	What would still close it fully
Propulsion in forward flight	Quantitatively bounded: margin survives up to 30% assumed thrust de-rating	A measured thrust-vs-airspeed curve for the chosen propeller
Complete-mission energy	Bounded estimate: about 47 Wh (no-tow) and 114 Wh (tow), both well inside budget	A single continuous simulation run covering launch to landing in one pass
Capture and recovery	Explicitly unvalidated; sizing geometry only	A deployment-dynamics model or physical net/parachute testing
Safe-area boundaries	Illustrative, now grounded against DGCA’s real no-fly-zone distances	An actual competition or reference site with approved dimensions

Hand this document, together with the earlier sizing calculations PDF and the architecture PDF, to the next simulation pass as the authoritative source for mass, propulsion, energy and boundary assumptions. Nothing in this document changes any dimension, mass, or thrust figure already in the sizing PDF — it only adds the forward-flight sensitivity check and the complete-mission energy bound on top of those existing numbers.